NODAL QUOTIENTS OF A FORMAL POWER SERIES RING CHARACTERIZED BY THE (N-1)-MINORS OF THE JACOBIAN MATRIX

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-curves/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the coordinate-and-generator change freedom from lemma-fitting-ideal-well-defined, the explicit verification of (2) and (3) in the nodal normal form, the unit-pivoting normalization loop, the block-shape estimate on the minors with the Nakayama and invertible-minor exclusions, and the final reduction to the nondegenerate quadratic term are all retained verbatim.

Editorial note. The label lemma-fitting-ideal names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Invariance of the Fitting-ideal criterion under coordinate and generator change. Same chapter, Lemma lemma-fitting-ideal-well-defined. Lets the proof freely change coordinates and regenerize the ideal I during the normalization steps, the pivot the whole argument rests on.
- The nodal quadratic model and reduced dimension-2 quotients. Same chapter, Lemmas lemma-nodal-algebraic and lemma-reduced-quotient-regular-ring-dim-2. Supplies side (1), the completed nodal quadric, and the structure theorem that an ideal in a two-variable power series ring defining a reduced quotient is generated by one element.

Source reference. Original reference 0C4B: sources/stacks-curves/pinned-original-source.tex, source lines 4282–4282. Prerequisite pointer invoked at proof line(s) 4328; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C49: sources/stacks-curves/pinned-original-source.tex, source lines 4090–4090. Prerequisite pointer invoked at proof line(s) 4317; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C48: sources/stacks-curves/pinned-original-source.tex, source lines 4034–4034. Prerequisite pointer invoked at proof line(s) 4382; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let $A = k[[x_1, \ldots, x_n]]$. Let $I = (f_1, \ldots, f_m) \subset \mathfrak{m}_A$ be an ideal. The following are equivalent*

1. *$k \to A/I$ is as in Lemma 0C49,*
2. *A/I is reduced and the $(n-1) \times (n-1)$ minors of the matrix $(\partial f_j/\partial x_i)$ generate $I + \mathfrak{m}_A$,*
3. *$\text{depth}(A/I) = 1$ and the $(n-1) \times (n-1)$ minors of the matrix $(\partial f_j/\partial x_i)$ generate $I + \mathfrak{m}_A$.*

Proof. By Lemma 0C4B we may change our system of coordinates and the choice of generators during the proof.

If (1) holds, then we may change coordinates such that $x_1, \ldots, x_{n-2}$ map to zero in A/I and $A/I = k[[x_{n-1}, x_n]]/(ax_{n-1}^2 + bx_{n-1}x_n + cx_n^2)$ for some nondegenerate quadric $ax_{n-1}^2 + bx_{n-1}x_n + cx_n^2$. Then we can explicitly compute to show that both (2) and (3) are true.

Assume the $(n-1)\times(n-1)$ minors of the matrix $(\partial f_j/\partial x_i)$ generate $I+\mathfrak{m}_A$. Suppose that for some i and j the partial derivative $\partial f_j/\partial x_i$ is a unit in A. Then we may use the system of parameters $f_j, x_1, \ldots, x_{i-1}, \hat{x}_i, x_{i+1}, \ldots, x_n$ and the generators $f_j, f_1, \ldots, f_{j-1}, \hat{f}_j, f_{j+1}, \ldots, f_m$ of I. Then we get a regular system of parameters $x_1, \ldots, x_n$ and generators $x_1, f_2, \ldots, f_m$ of I. Next, we look for an $i \geq 2$ and $j \geq 2$ such that $\partial f_j/\partial x_i$ is a unit in A. If such a pair exists, then we can make a replacement as above and assume that we have a regular system of parameters $x_1, \ldots, x_n$ and generators $x_1, x_2, f_3, \ldots, f_m$ of I. Continuing, in finitely many steps we reach the situation where we have a regular system of parameters $x_1, \ldots, x_n$ and generators $x_1, \ldots, x_t, f_{t+1}, \ldots, f_m$ of I such that $\partial f_j/\partial x_i \in \mathfrak{m}_A$ for all $i, j \geq t+1$.

In this case the matrix of partial derivatives has the following block shape

$$\begin{pmatrix} I_{t\times t} & * \\ 0 & \mathfrak{m}_A \end{pmatrix}$$

Hence every $(n-1) \times (n-1)$-minor is in $\mathfrak{m}_A^{n-1-t}$. Note that $I \neq \mathfrak{m}_A$ otherwise the ideal of minors would contain 1. It follows that $n - 1 - t \leq 1$ because there is an element of $\mathfrak{m}_A \setminus \mathfrak{m}_A^2 + I$ (otherwise $I = \mathfrak{m}_A$ by Nakayama). Thus $t \geq n - 2$. We have seen that $t \neq n$ above and similarly if $t = n - 1$, then there is an invertible $(n-1) \times (n-1)$-minor which is disallowed as well. Hence $t = n-2$. Then A/I is a quotient of $k[[x_{n-1}, x_n]]$ and Lemma 0C48 implies in both cases (2) and (3) that I is generated by $x_1, \ldots, x_{n-2}, f$ for some $f = f(x_{n-1}, x_n)$. In this case the condition on the minors exactly says that the quadratic term in f is nondegenerate, i.e., A/I is as in Lemma 0C49. □